Interquartile Range (IQR) and Outlier Analysis

This document compares the spread of the Day class and Night class data and identifies potential outliers using the 1.5 × IQR rule.

a. The interquartile range. Compare the two interquartile ranges.

b. Any outliers in either set.

The five number summary for the day and night classes is

	Minimum	Q_1	Median	Q_3	Maximum
Day	32	56	74.5	82.5	99
Night	25.5	78	81	89	98

a. Calculate the Interquartile Range (IQR)

The interquartile range (IQR) measures the spread of the middle 50% of a dataset.

Formula: IQR = $Q_3 - Q_1$

Class	Calculation	IQR
Day class	82.5 − 56	26.5
Night class	89 − 78	11

Comparison: The Day class has a larger IQR (26.5) than the Night class (11). Therefore, the middle 50% of the Day class data is more spread out, while the middle 50% of the Night class data is more concentrated.

b. Identify any outliers in either set

The 1.5 × IQR rule is used to identify potential outliers. Values below the lower bound or above the upper bound are considered potential outliers.

Lower bound = $Q_1 - 1.5 \times IQR$

Upper bound = $Q_3 + 1.5 \times IQR$

Day class

- IQR = 26.5
- Lesser Outlier = 56 − 1.5 × 26.5 = 16.25
- Greater Outlier = 82.5 + 1.5 × 26.5 = 122.25

The minimum value (32) and maximum value (99) are both within the ranges of 16.25 and 122.25.

Conclusion: There are no outliers in the Day class.

Night class

- IQR = 11
- Lesser Outlier = 78 − 1.5 × 11 = 61.5
- Greater Outlier = 89 + 1.5 × 11 = 105.5

The minimum value (25.5) is below the lesser Outlier (61.5), so it is a potential outlier. The maximum value (98) is within the ranges of 61.5 and 105.5.

Conclusion: The Night class has one potential outlier: 25.5.

Final Answers

1. Day class IQR: 26.5.
2. Night class IQR: 11.
3. The Day class has a greater spread in its middle 50% of values.
4. The Day class has no outliers.
5. The Night class has one potential outlier: 25.5.